Scheduled Flows in Crypto: Three Gold-Fix Mechanisms, None Tradable at Retail Cost

Oleksandr Sytnyk

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Abstract

Gold falls into its London fix because orders that must trade at a scheduled time move the price around that time. Crypto markets have their own scheduled obligations, and this note tests three of them on Bybit perpetual futures for bitcoin, ether and solana from 2020 to 2026. Funding payments every eight hours behave like an ex-dividend date: before settlement the price moves against the side that pays, by about half to two thirds of the payment ($t = -3.6$ across the three coins). The benchmark window used to value US spot bitcoin ETFs, 15:00–16:00 New York time, shows a dip into the window and a rebound after it since the ETFs launched in January 2024, and the rebound comes almost entirely on days of ETF outflows: +12 bp in the half hour after 16:00, $t = 5.0$, against −1 bp on inflow days. Deribit option expiries at 08:00 UTC on Fridays show the same dip and rebound, deepest before monthly expiries; across the three coins the half hour after expiry gains +8 bp ($t = 4.3$). All three mechanisms are real, and none pays: the moves are 1 to 12 bp, the round-trip taker cost is 11 bp, and the one conditional effect large enough to matter depends on ETF flows that are published only after the market has moved.

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1 Why look at crypto

Two companion notes found scheduled effects in precious metals: gold falls into the London AM auction, and gold and silver rise into the CME weekend close. In both, the explanation is the same: someone is obliged to trade at a known time, the obligation is one-sided on average, and the price moves to absorb it. Crypto has no auction of the same kind, but it has three schedules that create similar obligations:

- **Funding** on perpetual futures, paid every eight hours (00:00, 08:00 and 16:00 UTC on Bybit) from one side to the other. A position open at the settlement moment pays or receives the rate.
- **The ETF benchmark.** Several of the largest US spot bitcoin ETFs value their holdings at a CF Benchmarks reference rate computed from trades between 15:00 and 16:00 New York time. Creations and redemptions are settled at that value, so the bitcoin bought or sold for them is naturally traded in the window.

- **Option expiry** on Deribit, the largest crypto options venue: contracts settle at 08:00 UTC against an average price over the preceding 30 minutes; weekly options expire every Friday and the largest, monthly and quarterly, on the last Friday of the month.

The question for each is the one asked of gold: does the price move around the scheduled time in the direction the obligation predicts, and is the move large enough to trade?

2 Data and method

Exhibit 1. Data and costs

Item	Detail
Prices	Bybit USDT perpetual futures, 5-minute bars, UTC: bitcoin from March 2020, ether from March 2021, solana from October 2021, all to October 2026
Funding rates	Bybit funding history for the same contracts: 7,164 settlements for bitcoin, 6,100 for ether, 5,338 for solana; only 8-hour intervals are used (solana briefly settled every two hours)
ETF flows	daily net creations and redemptions of all US spot bitcoin ETFs, January 2024 – October 2026, 703 trading days (Farside Investors)
Costs	Bybit taker fee 0.055% per side, 11 bp per round trip; maker fee 0.02% per side, 4 bp

Returns are in basis points, measured between 5-minute bar closes. Each effect is compared with a placebo: the same windows at a time without the obligation (four hours from a funding settlement; the ETF window before the ETFs existed; Monday to Thursday for the Friday expiry). Mechanisms are judged by whether the effect changes in the direction the obligation predicts: with the sign of the funding rate, with the direction of ETF flows, with the size of the expiry.

3 Funding: an ex-dividend date every eight hours

A trader long a perpetual future when the funding rate is positive pays the rate at settlement; a trader who closes one minute before does not. If enough of the paying side closes before settlement, or the receiving side waits to open until just before it, the price should move against the payers before settlement by up to the size of the payment, as a share falls by its dividend on the ex-dividend date.

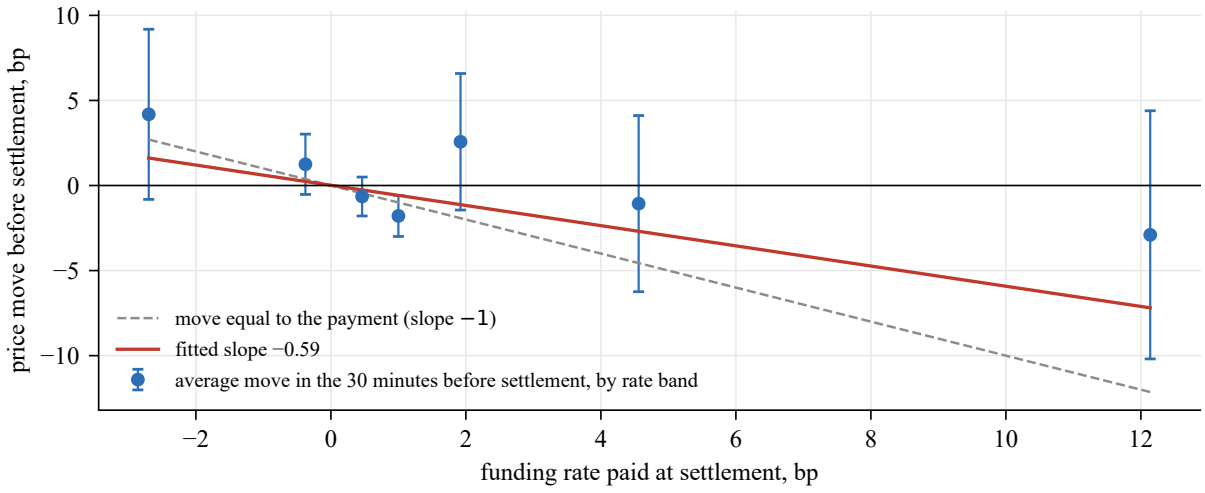
Exhibit 2. Price move before and after funding settlement

	Bitcoin	Ether	Solana	All three
Move against the paying side, 30 minutes before, bp	+1.38	+1.16	+1.51	–
t	+2.8	+1.8	+1.6	–
placebo: same window 4 hours later, bp	+0.18	−0.27	−0.87	–
Slope of the move before on the rate, bp per bp	−0.45	−0.80	−0.82	−0.60
t	−2.7	−1.9	−1.9	−3.6
Slope of the move after on the rate, bp per bp	–	–	–	+0.96
t	–	–	–	+5.2
Average funding rate, bp per settlement	1.16	0.80	0.39	–

Notes:
– Move against the paying side: the 30-minute move before settlement, signed so that a fall when longs pay (rate above zero) and a rise when shorts pay count as positive.
– Slope: regression of the move on the rate paid at that settlement; −1 would mean the price falls by exactly the payment. Settlements with rates beyond ±30 bp are excluded.
Source: Bybit 5-minute bars and funding history; author's calculations.

In all three coins the price moves against the payers before settlement, and the placebo four hours later does not. Across the three, the move before settlement falls by 0.60 bp for every basis point of funding: the price anticipates a little more than half of the payment (Exhibit 3). After settlement it moves the other way, and more so when the rate was high: settlements with a rate above 3 bp, 7.3 bp on average, are followed by +9.0 bp in the next half hour ($t = 3.6$).

Exhibit 3. Price move in the 30 minutes before funding settlement against the rate paid, three coins together



Notes:
– Points: average move for settlements grouped by rate, with 95% intervals. The largest group, two in five settlements, sits at the standard rate of 1 bp.
– The fitted line uses settlements with both windows available; the regression in Exhibit 2 uses all settlements with a move before.
Source: Bybit; author's calculations.

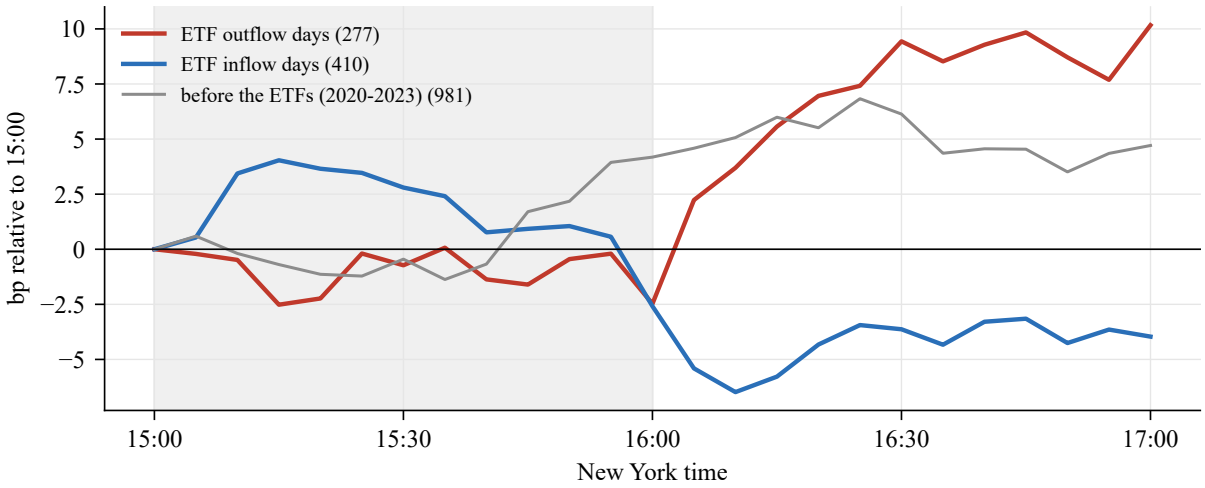
The effect is real but small. At the typical rate of 1 bp it is worth about 1 bp, and in bitcoin it

has faded from +3.5 bp in 2021 to almost nothing in 2025–2026. It also explains little: the points scatter widely around the line, and most of the variation in the half hour before settlement has nothing to do with funding.

4 The ETF benchmark window

Since the US spot bitcoin ETFs launched on 11 January 2024, bitcoin has tended to dip in the second half of the benchmark window and recover after it: −4.0 bp from 15:30 to 16:00 ($t = -2.7$) and +4.2 bp from 16:00 to 16:30 ($t = 2.8$). Before the ETFs the same half hour into 16:00 rose by +4.4 bp: the pattern appeared with them. The dip has faded, from −10.0 bp in 2024 to +2.4 bp in 2026, but the rebound has not.

Exhibit 4. Average bitcoin price from 15:00 to 17:00 New York time by ETF flow on the day, and before the ETFs



Notes:
– Bybit bitcoin perpetual, 5-minute closes relative to 15:00; weekdays; number of days in brackets.
– Shaded: the benchmark window, 15:00–16:00.
Source: Bybit; Farside Investors; author's calculations.

Exhibit 5. Bitcoin around the ETF benchmark window by the day's ETF flow, January 2024 – October 2026

Days	Number	Average flow, $m	15:30→16:00, bp	16:00→16:30, bp
All days since the ETF launch	703	–	−4.01 (t −2.7)	+4.20 (t +2.8)
Before the ETFs (2020–2023)	991	–	+4.43 (t +2.4)	+1.45 (t +0.9)
Net outflow days	277	–	−1.77 (t −0.7)	+11.94 (t +5.0)
Net inflow days	410	–	−5.37 (t −2.7)	−1.03 (t −0.5)
Largest outflows (fifth of days)	142	−351	–	+13.04
Largest inflows (fifth of days)	141	+580	–	+0.74
Previous day an outflow	276	–	−4.77 (t −1.9)	+4.61 (t +2.0)
Previous day an inflow	410	–	−3.11 (t −1.6)	+3.65 (t +1.8)

Notes:
– Weekdays with a published flow; 16 days with exactly zero flow are in neither group.
– Rebound after 16:00 on outflow days by year: +10.9 bp (2024), +13.9 bp (2025), +10.6 bp (2026); on inflow days −2.2, −1.5 and +1.7 bp.
Source: Bybit; Farside Investors; author's calculations.

The rebound belongs to outflow days. When the ETFs redeemed shares, bitcoin rose +11.9 bp in the half hour after the window ($t = 5.0$), in each of the three years; when they created shares, it did not move. Selling for redemptions, done in or around the benchmark window and released at 16:00, would produce exactly this. The window itself is less clear: the average path dips on both kinds of day, so the benchmark trades are not the only flow in that hour.

The useful condition, however, cannot be known in time. Daily ETF flows are published after the US close, and the previous day's flow, though correlated with today's (+0.52), does not separate the rebound: +4.6 bp after an outflow day against +3.7 bp after an inflow day.

5 Deribit option expiry

Deribit settles its options at 08:00 UTC against the average price of the preceding half hour. On Fridays, when the weekly options expire, all three coins dip into the averaging window and rebound after it; on Monday to Thursday, when only daily options expire, they do not (Exhibit 6). The rebound is positive in every year for all three coins, though barely in 2022.

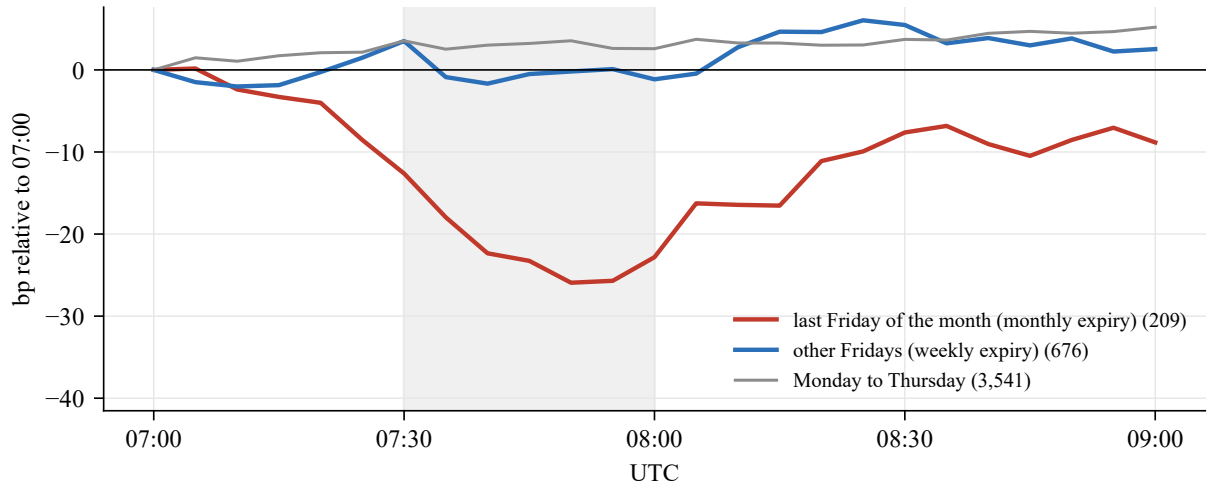
Exhibit 6. The half hour before and after 08:00 UTC, bp

Coin	Days	07:30→08:00	08:00→08:30
Bitcoin	Fridays (341)	−3.53 (t −1.7)	+6.01 (t +2.7)
	Monday–Thursday	−0.14	+1.82
Ether	Fridays (290)	−7.83 (t −2.7)	+6.75 (t +2.0)
	Monday–Thursday	−0.54	−0.79
Solana	Fridays (258)	−7.26 (t −1.8)	+11.93 (t +2.8)
	Monday–Thursday	−3.31	+1.99
All three	last Friday of the month (209)	−11.33 (t −3.0)	+7.58 (t +1.9)
	other Fridays (680)	−4.38 (t −2.3)	+8.08 (t +3.8)
	all Fridays (889)	−6.02 (t −3.5)	+7.96 (t +4.3)

Notes:
– Bitcoin Friday rebound by year: +5.6, +8.9, +0.1, +7.1, +4.6, +4.8 and +12.4 bp (2020–2026).
Source: Bybit; author's calculations.

The size of the expiry matters in the way the mechanism predicts: before the monthly and quarterly expiries on the last Friday of the month the dip is two and a half times deeper, −11.3 bp against −4.4 bp, and on average it starts well before the averaging window opens (Exhibit 7). Hedges of expiring options being unwound into the settlement average, and the price recovering once the average is fixed, fit this; the data do not show who trades.

Exhibit 7. Average price from 07:00 to 09:00 UTC, three coins together, by type of day



Notes:
– 5-minute closes relative to 07:00; coin-days in brackets. Shaded: the settlement average, 07:30–08:00.
Source: Bybit; author's calculations.

6 Can any of it be traded?

Exhibit 8. The simplest trade on each effect, after costs, bp per trade

Trade	Gross	Taker, 11 bp	Maker, 4 bp
Fade the paying side 30 minutes before funding (bitcoin)	+1.4	−9.6	−2.6
Long bitcoin 16:00→16:30 New York, every day	+4.2	−6.8	+0.2
only after a previous-day outflow	+4.6	−6.4	+0.6
only on outflow days (not knowable in time)	+11.9	+0.9	+7.9
Long 08:00→08:30 UTC on Fridays, three coins	+8.0	−3.0	+4.0

Notes:
– Round-trip costs on Bybit at the base fee level; no spread or slippage beyond the fee is assumed.
– The maker column assumes both orders fill as limit orders at the bar close, which a trade that needs the price to move away from the order rarely achieves.
Source: author's calculations.

None of the effects survives the taker fee. The Friday expiry trade comes closest and turns positive only at maker fees, +4.0 bp per trade with $t = 2.1$, but a limit order placed for a rebound fills most reliably when the rebound does not come, so the maker column overstates what is achievable. The ETF rebound on outflow days would pay even at taker cost, but the information that selects those days arrives too late.

7 Limitations

- **One venue.** All prices are Bybit perpetuals. Spot markets and other exchanges may show the effects more or less strongly, and the funding mechanism is specific to each venue's schedule.
- **Short ETF history.** The ETF results rest on 703 days, and the dip into the window has already faded within that period. The flow data come from a public aggregator, not from the issuers directly.
- **Daily expiries in the placebo.** Deribit options also expire at 08:00 on other days, so Monday to Thursday are days with small expiries, not none; the Friday contrast understates the full expiry effect.
- **Mechanisms inferred.** As in the gold notes, nobody's trades are observed; the mechanisms are supported by how the effects change with the obligation, not by position data.

8 Conclusion

The mechanism behind the London gold fix is not specific to gold. Wherever an obligation forces trading at a known time, crypto prices bend around it: they move against the payers before funding, dip into option expiry and recover after it, and rebound after the ETF benchmark on days the ETFs sell. What separates these effects from the gold fix is size relative to cost. The fix earns about 1.6 bp against a CFD cost of about 0.6 bp; the crypto effects earn 1 to 8 bp against a taker cost of 11 bp. For a researcher they confirm the mechanism in a second asset class; for a trader at retail fees they are something to know, not something to trade.